

An Elementary Approach to Serre's Problem on Projective Modules

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by
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CERTIFICATE

This is to certify that the project titled “**An Elementary Approach to Serre’s Problem on Projective Modules**” that is being submitted by Mr. Harikrishnan T R to Cochin University of Science and Technology, Cochin- 682022, is a bonafide record of study carried out by him under my supervision and guidance at the Department of Mathematics, Cochin University of Science and Technology during September 2023 - May 2023. This project has not been submitted anywhere for any other degree of any other University or Board.

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Declaration

I, Harikrishnan T R, hereby declare that this project report entitled “**An Elementary Approach to Serre’s Problem on Projective Modules**” that is being submitted to Cochin University of Science and Technology is a bonafide record of work done by me under the supervision of Dr. Ambily A. A. I also declare that this project has not been submitted for any other degree elsewhere.

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Chapter 1

INTRODUCTION

In this project report, we focus on the Serre's problem on projective modules, which was a longstanding and significant question in the realm of commutative algebra. The problem was first posed by Jean-Pierre Serre in 1955 and has since stimulated extensive research and investigation by numerous mathematicians. Serre suggested that finitely generated projective modules over a polynomial ring $k[x_1, \dots, x_n]$, where k is a field is free. Keeping this conjecture in view, we investigate the rings over which finitely generated projective modules and stably free modules are free. In 1976, Quillen and Suslin arrived at the first complete solutions of Serre's Conjecture independently of each other. In this project we aim to comprehend the solution of Serre's problem using a matrix theoretic approach developed by Mel Hochster by examining the structure of finitely generated projective modules.

In Chapter 2, we recall the preliminaries needed to proceed. Chapter 3 summarises the matrix theory over commutative rings. This chapter forms a preparatory tool for proceeding into the main theorem. This chapter focuses on Suslin's normality theorem which was proved when Suslin attempted a K_1 analogue of Serre's conjecture.

In Chapter 4, we discuss the theory of free modules and projective

modules. Here we investigate the rings over which projective modules are free. Projective modules are important algebraic objects in commutative algebra and K - theory. We also discuss the topology of a commutative ring in this chapter.

Chapter 5 describes the structure theory of stably free modules. Stably free modules form an important class of projective modules. There exist stably free modules which are not free. Therefore, these ‘stable’ objects are of particular interest and are also important in proving Serre’s theorem.

The final chapter, Chapter 6 discusses the solution of Serre’s problem. Here a matrix theoretic approach is employed to understand the solution. The solution follows from a series of theorems concerning completion of unimodular vectors and results on stably free projective modules.

Chapter 2

PRELIMINARIES

2.1 Rings

Definition 2.1.1. A non-empty set R together with two binary operations addition and multiplication is called a **ring** if; under addition, the set forms an abelian group and Multiplication is associative and distributive over addition.

$$\text{i.e., } (x + y)z = xz + yz. \quad x(y + z) = xy + xz.$$

A ring is said to have identity if there exist an element $1 \in R$ such that

$$1x = x1 = x \text{ for every } x \text{ in } R.$$

A ring is said to be commutative if multiplication is commutative.

We shall refer R as a commutative ring with 1, unless it is specified.

Definition 2.1.2. A subset I of ring R is an **ideal** if $x+y \in I$ and $ax \in I$ for all $x, y \in I$ and $a \in R$.

Definition 2.1.3. A proper ideal I of ring R is said to be **prime ideal** if $ab \in I$ implies that either $a \in I$ or $b \in I$.

An ideal I is said to be **maximal ideal** if there exist no other ideal J such that $I \subset J \subset R$.

Definition 2.1.4. Let A, B be two rings. A map $\phi : A \rightarrow B$ is a **ring homomorphism** if $\phi(a + b) = \phi(a) + \phi(b)$ and $\phi(ab) = \phi(a)\phi(b)$ for every a and b belonging to A .

Definition 2.1.5. An element of a ring A is called a **zerodivisor** if there exists $b \in A, b \neq 0$, such that $ab = 0$. A ring A is called an **integral domain** if $A \neq 0$ and every nonzero element of A is a nonzerodivisor.

Definition 2.1.6. An element $x \in A$ is said to be a **unit** if there exist an inverse element $y \in A$ such that $xy = 1$.

Definition 2.1.7. A ring A is called a **principal ideal domain (PID)** if A is an integral domain and every ideal of A is principal, that is, every ideal is generated by a single element.

Example 1. The ring $\mathbb{Z}$ of integers and the polynomial ring $k[X]$ in one variable over a field k are principal ideal domains.

If every non zero element in the ring is a unit, it is known as a field. An ideal I is prime ideal *iff* A/I is integral domain and an ideal is maximal ideal *iff* A/I is a field.

Definition 2.1.8. An element e of a ring A is called an **idempotent element** if $e^2 = e$. 0 and 1 are obviously idempotent elements in a ring with unity.

Definition 2.1.9. A ring A with exactly one maximal ideal m is called a **local ring**. The field A/m is called the residue field of A .

Definition 2.1.10. A ring is said to be **Noetherian** if it satisfies the ascending chain condition on ideals. *i.e.*, every increasing sequence of ideals $I_1 \subset I_2 \subset \dots$ has a largest element. It means that there exist a stage n such that $I_n = I_{n+1}$.

Definition 2.1.11. The **radical** of an ideal $\mathfrak{a}$ denoted by $\sqrt{\mathfrak{a}}$, defined by

$$\sqrt{\mathfrak{a}} = \{x \in A \mid x^n \in \mathfrak{a} \text{ for some positive integer } n.\}$$

When A is commutative, clearly $\sqrt{\mathfrak{a}}$ is an ideal of A .

Definition 2.1.12. The **Jacobson radical** of A , denoted by $\mathbf{r}(A)$, is the intersection of all maximal ideals of A . An element $a \in A$ is said to be nilpotent if $a^n = 0$ for some positive integer n . The set of all nilpotent elements of A forms an ideal of A called the **nilradical** of A denoted by $\text{nil } A$. In particular, $\text{nil } A$ is the intersection of all the prime ideals of A .

2.2 Module

Definition 2.2.1. Let R be a commutative ring. An R -**module** is a set M together with

- (i) a binary operation addition under which M is an abelian group and
- (ii) an action R on M satisfying
 - (a) $(x + y)z = xz + yz$ for every $x, y \in R$ and $z \in M$.
 - (b) $a(x + y) = ax + ay$ for every $a \in R$ and $x, y \in M$.
 - (c) $(ab)x = a(bx)$ for every $a, b \in R$ and $x \in M$.
 - (d) if ring has $1 \in R$ then $1x = x1 = x$ for every $m \in M$.

Definition 2.2.2. Let R be a ring and M, N be two R -modules. A map $\phi : M \rightarrow N$ is said to be an R -**module homomorphism** if it preserves the module structure. *i.e.*,

$$\phi(m + n) = \phi(m) + \phi(n); \quad \phi(am) = a\phi(m)$$

for every $m, n \in M, a \in R$.

Definition 2.2.3. A sequence of R -modules and R -module homomorphisms

$$\dots \rightarrow M_{i-1} \xrightarrow{f_i} M_i \xrightarrow{f_{i+1}} M_{i+1} \rightarrow \dots$$

is said to be **exact** at M_i if $\text{Im}(f_i) = \text{Ker}(f_{i+1})$. The sequence is exact if it is exact at each M_i .

In particular

$$0 \rightarrow M \xrightarrow{f} N \xrightarrow{g} P \rightarrow 0$$

is exact iff f is injective, g is surjective and g induces an isomorphism of $\text{Coker}(f) = N/f(M)$ onto P .

Example 2. A R -homomorphism $f: M \rightarrow N$ induces an exact sequence

$$0 \rightarrow \ker f \xrightarrow{j} M \xrightarrow{f} N \xrightarrow{\eta} \text{coker } f \rightarrow 0$$

where j and η are the natural inclusion and surjection, respectively.

Definition 2.2.4. An exact sequence $M \xrightarrow{\sigma} N \rightarrow 0$ **splits** if there is a homomorphism $\rho: N \rightarrow M$ with $\sigma\rho = \text{identity on } N$.

2.3 Group Actions

Definition 2.3.1. Let G be a group and E be a set. The group G is said to **act** on E if there exist a map $G \times E \rightarrow E$ sending $(g, e) \rightarrow ge$ satisfying

- (i) $1s = s$ for 1 (identity) in G and $s \in E$
- (ii) $(g_1g_2)s = g_1(g_2s)$ for $g_1, g_2 \in G$ and $s \in E$.

Let $s \in E$. Then **orbit** of s is the subset of E defined as

$$[s] = \{gs \mid g \in G, s \in E\}.$$

The orbits of elements of E partition E . If there is only one orbit of E under G , then G is said to be **transitive** on E .

Example 3. $\text{GL}_n(R)$ acts on $M_n(R)$ as:

$$\text{GL}_n(R) \times M_n(R) \rightarrow M_n(R)$$

by $(P, A) \rightarrow PAP^{-1}$. This action is called conjugation action.

2.4 Elementary Ideas

Let R be a commutative ring with 1. The set of all matrices over R of size $m \times n$ denoted by $M_{m \times n}(R)$ forms an R -module under usual matrix addition and scalar multiplication. If $m = n$, $M_{m \times n}(R)$ is abbreviated to $M_n(R)$.

Definition 2.4.1. The $n \times n$ matrix having 1 in the (i, j) - position and 0's in all other positions is called an **elementary matrix unit** denoted by $E_{i,j}$.

Since each element in $M_n(R)$ can be expressed uniquely as the finite linear combination of elementary matrix units, $M_n(R)$ forms a free R -module.

Definition 2.4.2. A matrix A in $M_n(R)$ is **invertible** if there is a matrix B with $AB = I$.

2.5 Ideals in $M_n(R)$

We can easily see that if A is an ideal in R , then

$$M_n(A) = \{[a_{ij}] \mid a_{ij} \text{ is in } A\}$$

is an ideal in $M_n(R)$.

Theorem 2.5.1. *Each ideal B of $M_n(R)$ has the form $B = M_n(A)$ for a unique ideal A of R .*

Proof. Let B be an ideal in $M_n(R)$. Let A denote the set of all elements r in R such that r occurs in the first row or column of some matrix in B . If r occurs in the first row (column) of a matrix X in B , then r occurs in the first row (column) of $E_{1i}X$ (XE_{i1}) of B where E_{1i} (E_{i1}) is an elementary matrix unit. Hence we may assume that all the elements of A can be found in the (1,1) th position of elements of B . Clearly A is an ideal of R .

Let $\bar{B} = M_n(A)$. If $\{E_{ij}\}$ are the elementary matrix units and $X = \sum x_{ij}E_{ij}$ is in $M_n(R)$, then

$$E_{pr}XE_{sq} = E_{pr} \sum x_{ij}E_{ij}E_{sq} = x_{rs}E_{pq}.$$

Thus we can isolate, by arithmetic with matrix units, elements of a given matrix and move them to new desired positions. Therefore $B \subset \bar{B}$. On the other hand, if $X = [X_{ij}]$ is in $\bar{B}$, then for each x_{ij} , there is a matrix $Y = [Y_{ij}]$ in B with $y_{11} = x_{ij}$. Then

$$E_{i1}YE_{1j} = y_{11}E_{ij} = x_{ij}E_{ij}$$

is in B . But then $X = \sum x_{ij}E_{ij}$ is in B . This completes the proof. ■

Corollary 2.5.1. *The map $A \rightarrow M_n(A)$ defines a bijection from the ideals of R to the ideals of $M_n(R)$ which is order preserving.*

Chapter 3

MATRIX THEORY OVER COMMUTATIVE RINGS

3.1 Determinant

Determinant is one of the most useful invariants associated with a matrix.

Definition 3.1.1. Let $A = [a_{ij}]$ be a matrix in $M_n(R)$. Then **determinant** of A is defined as

$$\det(A) = \sum_{\sigma} \operatorname{sgn}(\sigma) a_{1\sigma(1)} a_{2\sigma(2)} \cdots a_{n\sigma(n)}$$

where the summation extends over all permutations of $\{1, 2, \dots, n\}$.

Therefore, determinant is a continuous map from $M_n(R)$ to R .

Properties of Determinant

(i) $\det(A^T) = \det(A)$

(ii) If two columns of a matrix A are identical, then $\det(A) = 0$

Let $A \in M_{m \times n}(R)$. Let

$$A \begin{bmatrix} r_1 & r_2 & \cdots & r_l \\ c_1 & c_2 & \cdots & c_l \end{bmatrix} \quad l \leq \min\{n, m\}$$

denote the square $l \times l$ submatrix obtained from A by deleting all the rows and columns of A except the rows $r_1, r_2, \dots, r_l$ and the columns $c_1, c_2, \dots, c_l$.

Theorem 3.1.1. *Let A be an $m \times n$ matrix over R and B be an $n \times p$ matrix over R . Let*

$$M = \begin{bmatrix} r_1 & r_2 & \dots & r_l \\ c_1 & c_2 & \dots & c_l \end{bmatrix}$$

be any $l \times l$ submatrix of $M = AB$ where $1 \leq l \leq \min\{n, m, p\}$. Then

$$\det \left[M \begin{bmatrix} r_1 & r_2 & \dots & r_l \\ c_1 & c_2 & \dots & c_l \end{bmatrix} \right] = \sum \det \left[A \begin{bmatrix} r_1 & r_2 & \dots & r_l \\ c_1 & c_2 & \dots & c_l \end{bmatrix} \right] \det \left[B \begin{bmatrix} s_1 & s_2 & \dots & s_l \\ c_1 & c_2 & \dots & c_l \end{bmatrix} \right]$$

where the summation runs over all selections of $\{s_1, s_2, \dots, s_l\}$ from $\{1, 2, \dots, n\}$ with $1 \leq s_1 \leq s_2 \leq \dots \leq s_l$.

For proof see, ch.1. of [1].

Putting $m = n = p = l$ we obtain the corollary:

Corollary 3.1.1. *For A and B in $M_n(R)$, $\det (AB) = \det (A) \det (B)$.*

3.2 Method of Algebraic Identities

In this section we shall discuss how the properties of matrices over a field can be carried over to the case of matrices over an arbitrary commutative ring with 1. The idea is to prove statements in general rings by proving analogous statements for polynomials with integer coefficients.

For example, if we want to prove the identity

$$\det (AB) = \det (A) \det (B)$$

over commutative rings, we prove it for the variable matrices X, Y in the ring of integer polynomials $\mathbb{Z}[\{x_{ij}\}, \{y_{ij}\}]$ in $2n^2$ variable matrix entries. Since there is a unique homomorphism from $\mathbb{Z}$ to any ring R , given matrices $A = (a_{ij})$ and $B = (b_{kl})$ there is a homomorphism $\mathbb{Z}[\{x_{ij}\}, \{y_{kl}\}] \rightarrow R$, the substitution homomorphism which sends $x_{ij} \rightarrow a_{ij}$ and $y_{kl} \rightarrow b_{kl}$.

For an identity, all of whose terms are polynomials with integer coefficients in the matrix we must notice that the terms are compatible with ring homomorphisms. Therefore if our identity holds for the matrices in $\mathbb{Z}[\{x_{ij}\}, \{y_{kl}\}]$, then it also holds for the R -matrices A, B . Let K denote the field of quotient of the domain $\mathbb{Z}[\{x_{ij}\}, \{y_{kl}\}]$. Then since the corollary 3.1.1 holds over fields, it is valid for K and in particular, for the subring $\mathbb{Z}[\{x_{ij}\}, \{y_{kl}\}]$. Thus applying substitution homomorphism, we get equality in R .

Definition 3.2.1. Let $A \in M_n(R)$. Let A_{ij} denote the $(n-1) \times (n-1)$ matrix obtained from A by deleting the i -th row and j -th column. Set $b_{ij} = (-1)^{i+j} \det(A_{ij})$, $B = [b_{ij}]$ and $\text{adj}(A) = B^t$.

The $\text{adj}(A)$ is called the **adjoint** of A .

By the method of algebraic identities, we obtain the following theorem:

Theorem 3.2.1.

- (i) $A(\text{adj } A) = (\text{adj } (A))A = \det(A)I$
- (ii) $A^{-1} = (\det(A))^{-1} \text{adj}(A)$

3.3 General Linear Group

Definition 3.3.1. The set of all invertible matrices in $M_n(R)$ form a group under matrix multiplication called the **general linear group** over R of order n and is denoted by $\text{GL}_n(R)$.

Theorem 3.2.1 shows that the map $\det: M_n(R) \rightarrow R$ gives a surjective homomorphism $\det: GL_n(R) \rightarrow R^*$, where R^* denotes the units in R . The kernel of the homomorphism denoted by $SL_n(R) = \{A \in GL_n(R): \det(A)=1\}$ is called the special linear group of order n . It is clear that $A \in M_n(R)$ is invertible iff $\det(A) \in R$. There is a natural exact sequence

$$1 \rightarrow SL_n(R) \rightarrow GL_n(R) \xrightarrow{\det} R^* \rightarrow 1.$$

Definition 3.3.2. If I denotes the identity matrix and E_{ij} an elementary matrix unit, then $T_{ij}(\lambda) = I + \lambda E_{ij}$ where $i \neq j$ and $\lambda \in R$, is called an **elementary transvection**. The set

$$\{T_{ij}(\lambda) : i \neq j, 1 \leq i, j \leq n, \lambda \in R\}$$

generates a subgroup of $SL_n(R)$ denoted by $E_n(R)$ and called the **group of elementary transvections**.

Note that the group $GL_n(R)$ (resp., $E_n(R)$, $SL_n(R)$) can be identified with a subgroup of $GL_{n+1}(R)$ by $A \rightarrow \begin{bmatrix} A & 0 \\ 0 & 1 \end{bmatrix}$ where A belongs to $GL_n(R)$. Then, $GL(R) = \cup_{n \geq 1} GL_n(R)$ is called the **stable linear group**. Similarly we can define $SL(R)$ and $E(R)$.

Definition 3.3.3. If G is a group, then for every x, y belonging to G the **commutator** of x and y is defined as the element $xyx^{-1}y^{-1}$ denoted by $[x, y]$.

Definition 3.3.4. For a unit u in R and $1 \neq n$, we define an **elementary dialation**, $D_i(u)$ as the matrix, $D_i(u) = I + (u - 1)E_{ii}$.

Theorem 3.3.1. *Elementary transvections satisfy the following commutator relation:*

(i) Let $A = [a_{ij}]$ be in $GL_n(R)$ with $A^{-1} = [\alpha_{ij}]$. Then

$$[A, T_{ij}(\lambda)] = T_{ij}(\lambda) - \lambda^2 a_{ji} \sum_t \alpha_{ti} E_{tj} - \lambda \sum_{t,s} \alpha_{ti} a_{js} E_{ts}.$$

If α is the i -th column of A and β is the j -th row of A^{-1} , then $AT_{ij}(\lambda)A^{-1} = I + \lambda\alpha\beta$.

(ii) If i, j, k are distinct indices, then $[T_{ij}(\lambda), T_{jk}(\bar{\lambda})] = T_{ik}(\lambda\bar{\lambda})$.

Proof. (i) A simple matrix multiplication computation gives (i).

(ii) First note that $T_{ij}(\lambda)^{-1} = T_{ij}(-\lambda)$ and $E_{ij}E_{jk} = E_{ik}$. Therefore

$$[T_{ij}(\lambda), T_{jk}(\bar{\lambda})] = (I + \lambda E_{ij})(I + \bar{\lambda} E_{jk})(I - \lambda E_{ij})(I - \bar{\lambda} E_{jk}).$$

By a simple computation, we get that

$$[T_{ij}(\lambda), T_{jk}(\bar{\lambda})] = I + \lambda\bar{\lambda}E_{ik} = T_{ik}(\lambda\bar{\lambda}).$$

■

Since $E_n(R)$ is generated by the elementary transvections; as a consequence of the above theorem we obtain the following corollary:

Corollary 3.3.1. *If $n \geq 3$, then $E_n(R) = [E_n(R), E_n(R)]$.*

3.4 Suslin's Normality Theorem

Definition 3.4.1. An n -tuple $[a_1, \dots, a_n]$ with a_i is in R , $1 \leq i \leq n$, is said to be **unimodular** if the ideal generated by $\{a_1, \dots, a_n\}$ is R .

Suslin's normality theorem shows the normality of the group $E_n(R)$ in $GL_n(R)$ when $n \geq 3$. The proof of this theorem will follow from several lemmas due to David Wright.

Theorem 3.4.1. *The group $E_n(R)$ contains all matrices of the form*

$$\begin{bmatrix} 1 & * & \cdots & * \\ 0 & 1 & \cdots & * \\ \vdots & & \ddots & \\ 0 & \cdots & & 1 \end{bmatrix} \text{ and of the form } \begin{bmatrix} 1 & 0 & \cdots & 0 \\ * & 1 & & \\ \vdots & & \ddots & \vdots \\ * & * & \cdots & 1 \end{bmatrix}.$$

Proof. When $n = 1$, this is true. Assume the theorem holds for $n - 1$.

If $\alpha = [a_2, \dots, a_n]$, then

$$\begin{bmatrix} 1 & \alpha \\ 0 & I \end{bmatrix} = \prod_{i=2}^n T_{1i}(a_i)$$

is in $E_n(R)$. Further, if $\bar{A}$ is in $E_{n-1}(R)$, then

$$\begin{bmatrix} 1 & \alpha \\ 0 & \bar{A} \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & \bar{A} \end{bmatrix} \begin{bmatrix} 1 & \alpha \\ 0 & I \end{bmatrix}$$

is in $E_n(R)$. For the second statement, observe that $[T_{ij}(\lambda)]^t = [T_{ji}(\lambda)]$. Thus $E_n(R)^t \subseteq E_n(R)$. ■

Lemma 3.4.1. Let $\alpha = [a_1, \dots, a_n]^t$ and $\beta = [b_1, \dots, b_n]$. If one of the b_i is 0, and $\beta\alpha = \sum b_i a_i = 0$, then $I + \alpha\beta$ is in $E_n(R)$.

Proof. Assume without loss of generality that $b_n = 0$. Then

$$I + \alpha\beta = \begin{bmatrix} 1 & 0 & \cdots & a_1 \\ \vdots & 1 & & \vdots \\ \vdots & & \ddots & a_{n-1} \\ 0 & \cdots & & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & \cdots & \\ 0 & 1 & & \\ & & \ddots & \\ b_1 & \cdots & b_{n-1} & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & \cdots & -a_1 \\ \vdots & 1 & & \vdots \\ & & \ddots & -a_{n-1} \\ 0 & \cdots & & 1 \end{bmatrix} \\ \begin{bmatrix} 1 & 0 & \cdots & \\ 0 & 1 & & \\ & & \ddots & \\ -b_1 & \cdots & -b_{n-1} & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & \cdots & \\ 0 & 1 & & \\ & & \ddots & \\ a_n b_1 & \cdots & a_n b_{n-1} & 1 \end{bmatrix}.$$

Since by 3.4.1, each of the matrix in the right hand side of the equation is in $E_n(R)$, we have $I + \alpha\beta$ is in $E_n(R)$. $\blacksquare$

Lemma 3.4.2. Let $\alpha = [a_1, \dots, a_n]$, be unimodular and $\beta = [b_1, \dots, b_n]$. Suppose that $\beta\alpha^t = 0$. Then there exists a_{ij} in R such that $\beta = \sum_{i < j} a_{ij}(a_j e_i - a_i e_j)$ where e_i is the row $[0, 0, \dots, 1, 0, \dots, 0]$ with 1 in the i -th position and zeros elsewhere.

Proof. Since $\alpha = [a_1, \dots, a_n]$ is unimodular, the ideal $\langle a_1, \dots, a_n \rangle = \{\sum r_i a_i \mid r_i \in R\} = R$. Therefore we can find $\{s_1, \dots, s_n\}$ with $\sum s_i a_i = 1$. Then, for a fixed j ,

$$b_j = (\sum s_i a_i) b_j = s_j s_j b_j + \sum_{i \neq j} s_i a_i b_j = s_j [\sum_{i \neq j} (-b_i a_i)] + \sum_{i \neq j} s_i a_i b_j$$

since $\beta\alpha^t = \sum a_i b_i = 0$. Set $a_{ij} = s_j b_i - s_i b_j$. Then

$$\beta = \sum_{i < j} a_{ij}(a_j e_i - a_i e_j).$$

$\blacksquare$

Theorem 3.4.2. Suslin's normality theorem

If R is a commutative ring with unity and $n \geq 3$, then $E_n(R)$ is normal in $GL_n(R)$.

Proof. Suppose that A in $GL_n(R)$ and $T_{ij}(\lambda)$ is an elementary transvection. Then by theorem 3.3.1, $AT_{ij}(\lambda)A^{-1} = I + \lambda\alpha\beta$ where α is the i -th column and β is the j -th row of A^{-1} . Since $AA^{-1} = I$, we have $\beta\alpha = 0$ where $\beta = [b_1, \dots, b_n]$ and $\alpha = [a_1, \dots, a_n]^t$. By lemma 3.4.2, $\beta = \sum_{i < j} a_{ij}(a_j e_i - a_i e_j)$. Apply lemma 3.4.1 to $I + \lambda\alpha(a_j e_i - a_i e_j)$ (since $n \geq 3$) and conclude that this element is in $E_n(R)$. Then

$$I + \lambda\alpha\beta = \prod_{i < j} [I + \lambda\alpha(a_{ij}(a_j e_i - a_i e_j))]$$

is in $E_n(R)$. Thus $AT_{ij}(\lambda)A^{-1} \in E_n(R)$. Since elementary transvections generate $E_n(R)$, we get that $E_n(R)$ is normal in $GL_n(R)$. ■

Thus $E_n(R)$ is a large subgroup of $GL_n(R)$ which conditions the theory of any group G sandwiched between $E_n(R)$ and $GL_n(R)$. To prove a result from any group G , $E_n(R) \leq G \leq GL_n(R)$, prove the result first for $E_n(R)$ and “lift” to G .

But $n = 2$ is a special case. When $n = 2$, $E_2(R)$ need not be normal in $GL_2(R)$; if $n = 2$, there are insufficient numbers of off diagonal positions to use in performing algebraic operations.

Example 4. The matrix $\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix} \in E_2(R)$

Example 5. $E_2(R)$ need not be equal to $SL_2(R)$. Let $R = \mathbb{Z}[Y]$.

P.M. Cohn showed that the matrix $\alpha = \begin{bmatrix} 1 + 2Y & 4 \\ -Y^2 & 1 - 2Y \end{bmatrix} \notin E_2(R)$.

But obviously this is in $SL_2(R)$ since its determinant is 1 (See p.57 of [2]).

Example 6. $E_2(R)$ need not be normal in $GL_2(R)$. Consider α in the above example. Let $\beta = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} \in GL_2(R)$. But $\alpha\beta\alpha^{-1} \notin E_2(R)$ (see [2]). Therefore, $E_2(R)$ is not normal in $GL_2(R)$.

Chapter 4

FREE MODULES AND PROJECTIVE MODULES

4.1 Free Modules

A module is a generalization of the concept of vector spaces over a field to a ring.

Definition 4.1.1. Let R be a commutative ring and M be an R -module. A system $S = \{s_i\}_{i \in I}$ of M is said to be linearly independent if the condition $\sum_{i \in I} a_i s_i = 0$ with $a_i \in R$ and $a_i = 0$ for almost all i implies that $a_i = 0$ for every i . We say that S is a basis of M if S generates M as an R -module and is linearly independent over R . An R -module is said to be free if it has a basis.

Example 7. (i) The module 0 is free with empty set as a basis.

(ii) A commutative ring R is a free module with basis 1 .

(iii) R^n is an R -module with basis $\{e_1, \dots, e_n\}$ where each e_i is the n -tuple $(0, \dots, 1, \dots, 0)$ where 1 occurs only on the i -th position.

Theorem 4.1.1. Let M be an R -module. Then there is a free R -module F and a surjective homomorphism $\theta : F \rightarrow M$. Hence every R -module is a quotient of a free R -module.

Proof. Let $\lambda = M$. Consider

$$F = \{ \sigma : \lambda \rightarrow R \mid \sigma(a) = 0 \text{ for all but finitely many } a \text{ in } \lambda \}.$$

Then F is an R - module under the pointwise addition and pointwise scalar multiplication of maps. Let $b_m : M \rightarrow R$ be the map defined by $b_m(m) = 1$ and $b_m(\alpha) = 0, \alpha \neq m$. Then F is a free R - module with the basis $B = \{b_m \mid m \in M\}$. Define $\theta : F \rightarrow M$ as $\theta(b_m) = m$ on the basis and linearly extend to the whole F . Then this homomorphism is surjective. ■

Definition 4.1.2. We say that a ring R satisfies the **invariant basis property** (IBP) if, for any integers $s, t \geq 0$, $R^s \cong R^t$ (as right modules) implies $s = t$.

Definition 4.1.3. Let F be a free R - module. If every basis of F has the same cardinality say λ , then F is said to have dimension λ . If R has the property that all its free modules have dimension, then R is said to be a **dimensional ring**.

Example 8. A field is a dimensional ring.

Theorem 4.1.2. A commutative ring is a dimensional ring.

Proof. Let F be a free R - module with bases $B = \{b_i : 1 \leq i \leq p\}$ and $C = \{c_j : 1 \leq j \leq n\}$. Let m be a maximal ideal of R . Then we know that R/m is a field. Clearly F/mF is a free R/m module with bases $\phi B = \{\phi(b_i) : 1 \leq i \leq p\}$ and $\phi C = \{\phi(c_j) : 1 \leq j \leq p\}$ where $\phi : F \rightarrow F/mF$ is the canonical homomorphism. Then from example 2, it follows that $n = p$. ■

Example 9. Let K be a field and V a K - vector space of countably infinite dimension. Let $S = \text{End}_K(V)$. Consider S as a module over itself. Then $S \cong S \oplus \cdots \oplus S$ (n summands) for any positive integer n . See ch.2, appendix of [1]. This is an interesting example of a

module over a non-commutative ring which has bases of cardinality n for $n = 1, 2, 3, \dots$

Let $R\text{-mod}$ denote the category of R -modules. i.e., the collection of all R -modules together with the sets of R -homomorphisms $\text{Hom}_S(M, N)$ from an R -module M to an R -module N .

Definition 4.1.4. A **functor** F from a category $\bar{A}$ of modules to a category $\bar{B}$ of modules is a correspondence which

- (i) Associates to each module of M of $\bar{A}$ a module $F(M)$ in $\bar{B}$, written $F: M \rightarrow F(M)$.
- (ii) Associates to each homomorphism $\sigma: M \rightarrow N$ in $\bar{A}$, a homomorphism $F(\sigma): F(M) \rightarrow F(N)$ satisfying
 - (a) $F(\text{identity on } M) = \text{identity on } F(M)$;
 - (b) F preserves composition.

4.2 Projective Modules

Definition 4.2.1. An R -module P is called **projective** if there exists a module Q so that the direct sum $P \oplus Q$ is free.

Example 10. (i) Every free module is a projective module.

(ii) A finite abelian group G is not a projective $\mathbb{Z}$ -module, because it is not free. [If $g \in G$ and $n = |G|$, then $ng = 0$, so g can never be part of a basis.]

(iii) A projective module need not be free. For example if p and q are distinct primes, then $R = \mathbb{Z}_{pq} = \mathbb{Z}_p \oplus \mathbb{Z}_q$. But $\mathbb{Z}_p$ and $\mathbb{Z}_q$ are projective but not free R -modules.

(iv) If $e \in R$ is idempotent, then, $P = eR$, the subring generated by e is projective since $R = eR \oplus (1 - e)R$.

Theorem 4.2.1. *(Standard characterizations of a Projective Module)*

Let P be an R -module. The following are equivalent:

- (i) P is a direct summand of a free R -module.
- (ii) For every surjection $\theta: M \rightarrow N$ of R -modules and every map $\sigma: P \rightarrow N$ there exists a map $f: P \rightarrow M$ so that $\sigma = \theta f$.
- (iii) The functor $N \rightarrow \text{Hom}_R(P, N)$ is exact.
- (iv) Every short exact sequence $M \xrightarrow{\rho} P \rightarrow 0$ splits.

Proof. (i) implies (ii): Let P be a direct summand of a free R module say F . Then $F = P \oplus Q$ for some R -module Q . Let π be the natural projection to P . Since F is free, we can find a $f^*: F \rightarrow M$ (define f^* on a basis) such that $\theta f^* = \sigma \pi$. Take $f = f^*|_P$. Then we get $\sigma = \theta f$.

(ii) implies (iii): If P is any module, then the functor $\text{Hom}_R(P, N)$ is left exact; that is, whenever

$$0 \rightarrow L \xrightarrow{\beta} M \xrightarrow{\theta} N \rightarrow 0$$

is exact, then

$$0 \rightarrow \text{Hom}_R(P, L) \xrightarrow{\beta^*} \text{Hom}_R(P, M) \xrightarrow{\theta^*} \text{Hom}_R(P, N)$$

is exact, where $\beta^*(f) = \beta f$ and $\theta^*(g) = \theta g$. This is because $\beta^*(f) = 0$ implies $\beta(f(x)) = 0$ for every x . But then $f = 0$ by the injectivity of β . Similarly from the surjectivity of θ and exactness of the first sequence, we get $\text{Im}(\beta^*) = \text{Ker}(\theta^*)$. Thus it only remains to show that θ^* is surjective. Let $\sigma: P \rightarrow N$ be any module homomorphism. Then by (ii), there is a $f: P \rightarrow M$ with $\sigma = \theta f$. Then $\theta^*(f) = \theta f = \sigma$ and θ^* is surjective.

(iii) implies (iv): Suppose that $M \xrightarrow{\rho} P \rightarrow 0$ is exact. This sequence

expands to an exact sequence

$$0 \rightarrow K \rightarrow M \xrightarrow{\rho} P \rightarrow 0$$

where $K = \text{Ker}(\rho)$. By (iii),

$$0 \rightarrow \text{Hom}_R(P, K) \rightarrow \text{Hom}_R(P, M) \xrightarrow{\rho^*} \text{Hom}_R(P, P) \rightarrow 0$$

is exact. Thus there exists a $f : P \rightarrow M$ with $\rho^* = \text{identity on } P$. That is, $\rho f = \text{identity on } P$.

(iv) implies (i) : By 4.1.1, there is a surjective homomorphism ρ from a free R -module M to P . Then the sequence $M \xrightarrow{\rho} P \rightarrow 0$ is exact. Then by (iv), $M \cong P \oplus \text{Ker}(\rho)$. Therefore P is a direct summand of a free R -module. ■

Theorem 4.2.2. *A direct sum $P = \bigoplus_{\alpha} P_{\alpha}$ is projective iff each P_{α} is projective.*

Proof. We have the natural equivalence of functors

$$\text{Hom}_R(\bigoplus_{\alpha} P_{\alpha}, \cdot) \cong \prod_{\alpha} \text{Hom}_R(P_{\alpha}, \cdot)$$

Thus the left-hand-side is exact iff each $\text{Hom}_R(P_{\alpha}, \cdot)$ is exact. ■

Lemma 4.2.1. Nakayama Lemma Let $\mathfrak{a}$ be an ideal contained in Jacobson radical of ring R , and let M be an R -module. If M is a finitely generated R -module and $\mathfrak{a}M = M$ then $M = 0$.

Proof. Let $x_1, x_2, \dots, x_n$ generate M . Choose the least integer n with this property. Assume that $n \geq 1$. Since $x_n \in \mathfrak{a}M$, $x_n = a_1x_1 + \dots + a_nx_n$ where $a_i \in R$. Then we get, $(1 - a_n)x_n = a_1x_1 + \dots + a_{n-1}x_{n-1}$. Now since, a_n is in Jacobson ideal, $1 - a_n$ is a unit. This means that $x_n \in \text{span}\{x_1, x_2, \dots, x_{n-1}\}$ which contradicts the minimality of n . Therefore, $n = 0$. i.e, $M = 0$. ■

Theorem 4.2.3. *Let R be a commutative ring and P a finitely generated projective R -module. Let $\mathfrak{a}$ be an ideal in the Jacobson radical $\text{Rad}(R)$ of R . If $P/\mathfrak{a}P$ is an $R/\mathfrak{a}$ -free module, then P is R -free. In particular, a finitely generated projective module over a local ring is free.*

Proof. Let $\mathfrak{a}$ be in $\text{Rad}(R)$. Since $P/\mathfrak{a}P$ is free, we have

$\Pi : P \rightarrow P/\mathfrak{a}P$, $\bar{\Pi} : R^n \rightarrow (R/\mathfrak{a})^n$ and $\bar{\sigma} : P/\mathfrak{a}P \rightarrow (R/\mathfrak{a})^n$ where Π and $\bar{\Pi}$ are natural homomorphisms and $\bar{\sigma}$ is an isomorphism. Since P is projective and $R^n \xrightarrow{\bar{\Pi}} (R/\mathfrak{a})^n \rightarrow 0$ is exact, there is a homomorphism $\sigma : P \rightarrow R^n$ such that the diagram

$$\begin{array}{ccc} P & \xrightarrow{\sigma} & R^n \\ \downarrow \Pi & & \downarrow \bar{\Pi} \\ P/\mathfrak{a}P & \xrightarrow{\bar{\sigma}} & (R/\mathfrak{a})^n \end{array}$$

is commutative. Since $\bar{\sigma}$ is surjective, $\text{Im}(\sigma) + \mathfrak{a}R^n = R^n$. Thus, by Nakayama's lemma, $\text{Im}(\sigma) = R^n$ and σ is surjective. Hence

$P \xrightarrow{\sigma} R^n \rightarrow 0$ is exact and, consequently splits with $P \cong R^n \oplus \ker(\sigma)$.

Thus $\ker(\sigma)$ is a finitely generated R -module. Since $\bar{\sigma}$ is injective, $\ker(\sigma)/\mathfrak{a}\ker(\sigma) = 0$. Thus $\ker(\sigma) = \mathfrak{a}\ker(\sigma)$ and, again by Nakayama's lemma, $\ker(\sigma) = 0$. Hence $\sigma : P \rightarrow R^n$ is an isomorphism. Therefore P is R -free. If R is a local ring, then R has only one maximal ideal say $\mathfrak{m}$ and then $\text{Rad}(R) = \mathfrak{m}$. Take $\mathfrak{a} = \mathfrak{m}$. Then $P/\mathfrak{a}P$ is an $R/\mathfrak{a}$ -free module since $R/\mathfrak{m}$ is a field and thus $P/\mathfrak{a}P$ is a vector space. Therefore P is R -free when R is a local ring. ■

4.3 Topology of a Commutative Ring

Definition 4.3.1. Let R be a ring. Set of all prime ideals in R is known as **spectrum of the ring**.

$$\text{Spec } R = \{\mathfrak{p} \mid \mathfrak{p} \text{ is prime ideal in } R\}.$$

Let $V(E) = \{\mathfrak{p} \in \text{Spec } R \mid E \subseteq \mathfrak{p}\}$. It is clear that $V(A) = \emptyset$, $V(0) = \text{Spec } R$, $V(\mathfrak{a}) \cup V(\mathfrak{b}) = V(\mathfrak{a}\mathfrak{b})$ for all ideals $\mathfrak{a}, \mathfrak{b}$ of R , and $\bigcap_{i \in I} V(\mathfrak{a}_i) = V(\sum_{i \in I} \mathfrak{a}_i)$ for every family $\{\mathfrak{a}_i\}_{i \in I}$ of ideals of R . Then, there is a topology on $\text{Spec } R$ with $V(\mathfrak{a})$ as closed sets as $\mathfrak{a}$ varies over ideals of R , called the **Zariski topology**.

Suppose e is a nontrivial idempotent of R . Then $(1 - e)$ is also an idempotent and $R = Re \oplus R(1 - e)$ where Re is the subring of R generated by the single element e .

Definition 4.3.2. A commutative ring is called **connected** if its prime spectrum is a connected topological space.

Theorem 4.3.1. A commutative ring R is connected if and only if its only idempotents are 0 and 1.

Proof. Assume R is connected. Suppose there exists an idempotent element e which is not 0 or 1. Then clearly $R = eR \oplus (1 - e)R$, so that $\text{Spec } R = V(e) \cup V(1 - e)$, where $V(e)$ and $V(1 - e)$ are closed disjoint sets. Therefore $\text{Spec } R$ is disconnected.

Suppose $\text{Spec } R$ is not connected. Then there exist ideals A and B of ring R such that $V(A) \cup V(B) = \text{Spec } R$ and $V(A) \cap V(B) = \emptyset$ with $V(A), V(B) \neq \emptyset$ or $\text{Spec } R$. It is easy to check the following:

1. $V(A) \cup V(B) = V(A \cap B) = \text{Spec } R$
2. $V(A) \cap V(B) = V(A + B) = \emptyset$
3. $A + B = R$ (This follows from 2)

4. $AB = A \cap B$ (This follows from 3)

5. $V(AB) = V(A \cap B) = \text{Spec } R$

Note that 5 implies that AB is contained in every prime ideal of R and hence every element of AB is nilpotent. Hence, for any $a \in A$ and for any $b \in B$, there exists a natural number n such that $(ab)^n = 0$. Thus $1 = (a + b)^n = a^n + b^n + abc$, for some $c \in R$. This gives $a^n + b^n = 1 - abc$ and hence is the sum of unit and a nilpotent element and, as such $a^n + b^n$ must be a unit. Hence there exists some $u \in R$ such that $u(a^n + b^n) = 1$, which yields $ua^n = ua^n(u(a^n + b^n)) = u^2a^{2n} + u^2(ab)^n = u^2a^{2n}$, proving ua^n is an idempotent element. In an analogous manner, we obtain ub^n is also idempotent. Finally, we must show $ua^n \neq 0$ or 1 . If $ua^n = 1$, then a is invertible which implies $A = R$ and thus $V(A) = V(R) = \emptyset$, a contradiction. Now if $ua^n = 0$, $a^n = 0$ so $1 = u(a^n + b^n) = ub^n$. Therefore b is invertible and so $V(B) = \emptyset$. This is a contradiction using the same argument as in case of A . Thus, $e = ua^n$ is our desired nontrivial idempotent if $\text{Spec } R$ is not connected. ■

Definition 4.3.3. A commutative ring R is said to be **projectively trivial** if each idempotent matrix over R is diagonalizable under a matrix conjugation action.

Theorem 4.3.2. *Let R be a projectively trivial commutative ring and P be a non-zero finitely generated projective module. If R is connected, then P is free.*

Proof. Let R be a projectively trivial ring and P a nonzero finitely generated projective R -module. Let F be free of finite dimension, say $\dim(F) = n$, over R with $P \oplus Q = F$. Let $\Pi : F \rightarrow F$ be the projection of F onto P and $A = \text{Mat}(\Pi)$ the matrix of the transformation with respect to some fixed basis.

Then $A^2 = A$ and A is an idempotent matrix. Since R is projectively trivial, there exists an invertible matrix Q with $Q A Q^{-1} = \text{diag}[e_1, e_2, \dots, e_t, 0, \dots, 0]$ where $1 \leq t \leq n$ and $e_i^2 = e_i \neq 0$, $1 \leq t \leq n$. Then,

$$P = \text{Im}(\Pi) \cong \text{Im}(A) \cong \text{Im}(Q A Q^{-1}) \cong R e_1 \oplus \dots \oplus R e_t.$$

In particular if R is connected, R has only trivial idempotent elements, then it is clear that P is free. ■

Chapter 5

STABLY FREE MODULES

5.1 Stably Free Modules

Definition 5.1.1. An R -module P is called **stably free** (of rank $n - m$) if $P \oplus R^m \cong R^n$ for some non-negative integers m and n .

Stably free modules form an important class of projective modules. Stably free modules need not be free. But all stably free modules are projective.

Example 11. The R -module $T = \{(f, g, h) \in R^3 : xf + yg + zh = 0 \text{ in } R\}$, where R is the ring $\mathbb{R}[x, y, z]/(x^2 + y^2 + z^2 - 1)$ is stably free, but not free (refer [8]).

Definition 5.1.2. Let $\alpha = [r_1, \dots, r_n]$ be a **unimodular vector** in R^n : i.e., the ideal generated by α is R . We say that α is **completable** if there is an $n \times n$ invertible matrix in $M_n(R)$ whose first row is $[r_1, \dots, r_n]$.

Theorem 5.1.1. *For a commutative ring R , the following conditions are equivalent.*

- (i) *For all R -modules M , if $M \oplus R \cong R^n$ for some n then M is free.*
- (ii) *All stably free finitely generated R -modules are free.*

(iii) For each positive integer m , every unimodular vector in R^m is completable.

Proof. (i) implies (ii): Suppose that M is stably free R -module, so $M \oplus R^k \cong R^l$ for some k and l . If $k = 0$, then obviously M is free. If $(M \oplus R^{k-1}) \oplus R \cong R^l$, so (i) with $n = l$ tells us that $M \oplus R^{k-1}$ is free. By induction on k , the module M is free.

(ii) implies (i): If $M \oplus R \cong R^n$ for some n then M is stably free and thus M is free by (ii).

(ii) implies (iii): Let $\alpha = [r_1, \dots, r_m]$ be a unimodular vector in R^m . Then $R^m = R\alpha \oplus P$ where $P = R^{m-1}$. But P is clearly stably free. Hence by (ii), P is free. Let $\alpha = \alpha_1$ and let $\alpha_2, \dots, \alpha_m$ be a basis for P . Then the matrix whose rows are $\alpha_1, \alpha_2, \dots, \alpha_m$ is invertible. Thus α is completable. ■

Theorem 5.1.2. *All stably free finitely generated R -modules are free when R is a field or PID.*

Proof. This holds when R is a field since all vector spaces have bases. This also when R is a PID: a stably free module over a PID is a submodule of a finite free R -module, and every submodule of a finite free module over a PID is a free module. ■

Theorem 5.1.3. *Let P be a projective R -module. Then there is a free R -module F of infinite dimension such that $P \oplus F$ is free.*

Proof. Since P is free there exists an R -module Q such that $P \oplus Q = \bar{F}$ for some free R -module $\bar{F}$. Then,

$$\begin{aligned} P \oplus \bar{F} \oplus \bar{F} \oplus \dots &\cong P \oplus (P \oplus Q) \oplus (P \oplus Q) \oplus \dots \\ &\cong P \oplus (Q \oplus P) \oplus (Q \oplus P) \oplus \dots \\ &\cong (P \oplus Q) \oplus (P \oplus Q) \oplus \dots \\ &\cong \bar{F} \oplus \bar{F} \oplus \dots \end{aligned}$$

Thus, if $F = \bar{F} \oplus \bar{F} \oplus \dots$, then $P \oplus F \cong F$. ■

Theorem 5.1.4. (Gabel) *If M is stably free and not finitely generated, then M is free.*

Proof. Let $F \cong M \oplus R^k$ be free. Projection of F onto M is a surjective linear map, so M is not finitely generated implies F is not finitely generated. Let $\{e_i\}_{i \in I}$ be a basis of F , so the index I is infinite. Projection of F onto R^k is a surjective linear map with kernel M . The standard basis of R^k is in the image of the span of finitely e_i 's, say the submodule $F' = Re_1 + \dots + Re_l$ has $f(F') = R^k$. For each $v \in F$, $f(v) = f(v')$ for some $v' \in F'$. Then $v - v' \in \ker f = M$, so $F = F' + M$. The module F' is finite free and $F'/(M \cap F') \cong R^k$. Since R^k is free, there is an isomorphism $F' \cong N \oplus R^k$ where $N = M \cap F'$. Since,

$$F' + M = F, \quad F/F' \cong M/N$$

and $F/F' = \bigoplus_{i > l} Re_i$ is free with infinite rank, so we can write $F/F' \cong R^k \oplus F''$ for some free F'' . Therefore M/N is free, so

$$M \cong N \oplus (M/N) \cong N \oplus (F/F') \cong N \oplus R^k \oplus F'' \cong F' \oplus F''$$

which is free. ■

Chapter 6

SERRE'S PROBLEM ON PROJECTIVE MODULES

6.1 A Brief Note on Serre's Problem on Projective Modules

Serre's problem was a significant open question in algebraic geometry and homological algebra that was formulated by French mathematician Jean-Pierre Serre. The problem seeks to understand the structure and classification of projective modules over polynomial rings. Serre's problem originated in the context of algebraic geometry, where he was investigating the notion of vector bundles over projective algebraic varieties. Serre was interested in understanding whether all projective modules over polynomial rings are free, a property known as projective freeness. In 1955, he suggested that finitely generated projective modules over a polynomial ring $k[x_1, \dots, x_n]$, where k is a field, is free. After 20 years of extensive research by numerous mathematicians, in 1976, Quillen and Suslin arrived at the first complete solutions of Serre's Conjecture in full generality.

In this chapter, we adopt a matrix theoretic approach to arrive at the solution. The solution follows from two major results; the first being that unimodular vectors over $k[x_1, \dots, x_n]$ are completable

to invertible matrices, and the second being that finitely generated projective modules over $k[x_1, \dots, x_n]$ have free complements, i.e., are stably free.

6.2 Completion of Unimodular Vectors

Let $M(R) = M_{m \times n}(R)$ denote the $m \times n$ matrix over a commutative ring R . Suppose that $B[X]$ is a matrix in $M(R[X])$. Then $B[X]$ can be written as

$$B = A_0 + A_1X + \dots + A_tX^t$$

for some A_i in $M(R)$. If $B[X]$ is in $M(R[X])$ and a is in R , then by $B[a]$ we mean that the image of $B[X]$ under the natural extension of the substitution homomorphism

$$R[X] \rightarrow R(X \rightarrow a) \rightarrow M(R[X]).$$

Let $G_1(R) = GL_n(R)$ and $G_2(R) = GL_m(R) \times GL_n(R)$. Then $G_1(R)$ acts on $M(R)$ by $g(x) = xg^{-1}$ for $x \in M(R)$, $g \in G_1(R)$ and $G_2(R)$ acts on $M(R)$ by $(g_1, g_2)(x) = g_1xg_2^{-1}$.

Theorem 6.2.1. *Let $B[X]$ be in $M(R[X])$*

1. *The following are equivalent:*

- (i) *The elements $B[X]$ and $B[0]$ lie in the same $G_1(R[X])$ orbit.*
- (ii) *The images of $B[X]$ and $B[0]$ in $M(R_q[X])$ lie in the same $G_1(R_q[X])$ orbit for each maximal ideal q of R .*

2. *The following are equivalent:*

- (a) *The elements $B[X]$ and $B[0]$ lie in the same $G_2(R[X])$ orbit.*
- (b) *The images of $B[X]$ and $B[0]$ in lie in the same $G_2(R_q[X])$ orbit for each maximal ideal q of R (p.340 [1]).*

Lemma 6.2.1. There is a change of variables

$$\begin{aligned} X_1 &\rightarrow Z = X_1 \\ X_2 &\rightarrow Y_1 = X_2 + X_1^{r_2} \\ &\vdots \qquad \qquad \qquad \vdots \quad \vdots \\ X_n &\rightarrow Y_{n-1} = X_n + X_1^{r_n} \end{aligned}$$

such that f_1 , viewed as a polynomial in $S[Z]$ where $S = k[Y_1, \dots, Y_{n-1}]$ has leading coefficient a unit in k .

Proof. Let $f_1(X_1, \dots, X_n) = \sum_i a_i X_1^{i_1} X_2^{i_2} \dots X_n^{i_n}$. Then

$$\begin{aligned} f_1(Z, Y_1, \dots, Y_{n-1}) &= f_1(X_1, X_2 + X_1^{r_1}, \dots, X_n + X_1^{r_n}) \\ &= \sum_i a_i X_1^{i_1} (X_2 + X_1^{r_2})^{i_2} \dots (X_n + X_1^{r_n})^{i_n} \end{aligned}$$

$f_1(Z, Y_1, \dots, Y_{n-1}) = \sum_i (a_i X_1^{i_1+r_2 i_2+\dots+r_n i_n}) + \text{expression whose degree in } X_1 \text{ is less than } (i_1 + r_2 i_2 + \dots + r_n i_n).$

Let m be an integer greater than $\max \{i_1, \dots, i_n\}$ for all i . Select $r_i = m^{i-1}$. Then the monomials $a_i X_1^{i_1+r_2 i_2+\dots+r_n i_n}$ will not cancel each other and the term with the highest degree and nonzero coefficient will be the leading term of f_1 as a polynomial in Z . This completes the proof. ■

Lemma 6.2.2. Let $R = S[X]$ where S is a local ring. Let f_1 be a polynomial in R whose leading coefficient is a unit. Consider a unimodular row $[f_1, \dots, f_t]$ over R . Then $[f_1, \dots, f_t]$ is in the same orbit as $[f_1(0), \dots, f_t(0)]$ under $\text{GL}_t(R)$.

Proof. Consider the ring $S[X]/\langle f_1 \rangle$. Since f_1 has leading coefficient a unit and S is local, $S[X]/\langle f_1 \rangle$ is semilocal (a commutative Noetherian unit ring having only finitely many maximal ideals).

Let $\bar{f}_i$ denote the images of the polynomials f_i in $S[X]/\langle f_1 \rangle$. But

$S[X]/\langle f_1 \rangle$ is semilocal implies that there is a product of elementary transvections in $\mathrm{GL}_t(S[X]/\langle f_1 \rangle)$ which will carry $[0, \bar{f}_2, \bar{f}_3, \dots, \bar{f}_t]$ to $[0, \bar{1}, 0, \dots, 0]$. Since elementary transvections in $\mathrm{GL}_t(S[X]/\langle f_1 \rangle)$ lift naturally to elementary transvections in $\mathrm{GL}_t(S[X])$, there is a product of elementary transvections in $\mathrm{GL}_t(S[X])$ that will carry $[f_1, \dots, f_t]$ to $[f_1, 1 + g_2 f_1, \dots, g_t f_t]$. Additional elementary transvections will transform $[f_1, 1 + g_2 f_1, \dots, g_t f_t]$ to $[f_1, 1, 0, \dots, 0]$ and then by $\mathrm{GL}_t(S[X])$ to $[1, 0, 0, \dots, 0]$. ■

Theorem 6.2.2. *Let k be a field. If $\alpha = [f_1, \dots, f_t]$ is a unimodular vector over $k[X_1, \dots, X_n]$, then α is completable to a matrix in $\mathrm{GL}_t(k[X_1, \dots, X_n])$.*

Proof. In lemma 6.2.1, we can assume f_1 as a polynomial in $S[X_1]$ where $S = k[X_2, X_3, \dots, X_n]$, has leading coefficient in a unit in k . Then by theorem 6.2.1 and lemma 6.2.2, we have $\alpha = [f_1(X_1, \dots, X_n), \dots, f_t(X_1, \dots, X_n)]$ is in the same orbit under $\mathrm{GL}_t(R)$ as the unimodular vector $\beta = [f_1(0, X_2, \dots, X_n), \dots, f_t(0, X_2, \dots, X_n)]$. Then by induction on n we may assume that α lies in the same orbit as $[1, 0, \dots, 0]$. Since α is an arbitrary unimodular vector, this will show that $\mathrm{GL}_t(R)$ acts transitively on unimodular vectors in R^t . Hence by theorem 5.1.1, α is completable. ■

6.3 Free Complements and $K_0(R)$

Suppose R is a commutative ring. Let P be a finitely generated R -module. Let $[P]$ denote the isomorphism class of P . The additive group $K_0(R)$ is the group generated by $[P]$ for each finitely generated projective module P and whose relations are $[P] = [P_1] + [P_2]$ whenever $0 \rightarrow P_1 \rightarrow P \rightarrow P_2 \rightarrow 0$ is an exact sequence of finitely generated projective R -modules; equivalently, the relations are:

$$(a) [P] = [\bar{P}] \text{ if } P \cong \bar{P}$$

$$(b) [P \oplus \bar{P}] = [P] + [\bar{P}]$$

Theorem 6.3.1. *The finitely generated projective module P is stably free iff $[P]$ is in $\langle [R] \rangle$ where $\langle [R] \rangle$ is the cyclic subgroup of $K_0(R)$ generated by $[R]$.*

Proof. If $[P]$ is in the cyclic subgroup $\langle [R] \rangle$ generated by $[R]$, then for some $m \geq 0$,

$$[P] = \sum_{i=1}^m [R] = m[R] = [R^m].$$

Then $P \oplus R^t \cong R^m \oplus R^t$ for some $t \geq 0$. Conversely, if P is stably free, then $P \oplus R^m \cong R^n$ for some m and n . Then $[P] = (n - m)[R]$ is in $\langle [R] \rangle$. ■

Definition 6.3.1. A commutative ring R is called **regular** if

- (i) R is a Noetherian ring.
- (ii) For any finitely generated R - module M , there is an exact sequence

$$0 \rightarrow P_n \rightarrow \dots \rightarrow P_0 \rightarrow M \rightarrow 0$$

of finitely generated projective R - modules.

Theorem 6.3.2. *(Swan's theorem [1] p.355). Suppose that R is a regular and M is a finitely generated $R[X]$ - module. Then there exist finitely generated projective R - module $P_0, P_1, \dots, P_n$ such that*

$$0 \rightarrow P_n[X] \rightarrow \dots \rightarrow P_1[X] \rightarrow P_0[X] \rightarrow M \rightarrow 0$$

is exact. In particular, $R[X]$ is regular.

Theorem 6.3.3. *Let R be a commutative regular ring. Then*

$$(i) K_0(R) \cong K_0(R[X_1, \dots, X_n])$$

(ii) If all finitely generated projective R -modules are stably free, then all finitely generated projective $R[X_1, \dots, X_n]$ -modules are stably free.

Proof. Suppose that R is regular and P is a finitely generated projective $R[X]$ -module. By Swan's theorem, there exist finitely generated projective R -modules $P_0, P_1, \dots, P_n$ such that

$$0 \rightarrow P_n[X] \rightarrow \dots \rightarrow P_1[X] \rightarrow P_0[X] \rightarrow P \rightarrow 0$$

exact. Then, in $K_0(R[X])$ we have $[P] = \sum_i (-1)^i [P_i[X]]$.

Thus the generators of $K_0(R)$ extend to generators of $K_0(R[X])$. That is, the natural injection $\Pi : K_0(R) \rightarrow K_0(R[X])$ is surjective. Hence $K_0(R) \cong K_0(R[X])$.

Additionally, if each finitely generated projective R -module is stably free, then by 6.3.1, $K_0(R) = \langle R \rangle$. In this case, $K_0(R[X]) = \langle R[X] \rangle$ and consequently every finitely generated projective $R[X]$ -module is stably free. ■

Corollary 6.3.1. *Let k be a field. Then every finitely generated projective $k[X_1, \dots, X_n]$ -module is stably free.*

Proof. Since every field has only two ideals; 0 and itself, k is Noetherian. Let M be a finitely generated k -module, that is a finite dimensional vector space. In example 2, taking $M = N$ and $f = 0$, we get an exact sequence of the form

$$0 \rightarrow P_n \rightarrow \dots \rightarrow P_0 \rightarrow M \rightarrow 0$$

where each P_i is a k -vector space. Therefore k is regular. Therefore by theorem 6.3.3 every finitely generated projective $k[X_1, \dots, X_n]$ -module is stably free. ■

Theorem 6.3.4. (*Serre's theorem on projective modules*) *Let k be a field. Then every finitely generated projective $k[X_1, \dots, X_n]$ -module is free.*

Proof. Let P be a finitely generated projective $k[X_1, \dots, X_n]$ -module. Then by corollary 6.3.1, we have P is stably free, i.e., there exist non-negative integers m and n such that $R^n \cong P \oplus R^m$ where $R = k[X_1, \dots, X_n]$. By theorem 6.2.2, we have unimodular vectors over $k[X_1, \dots, X_n]$ are completable to invertible matrices.

Suppose that $M \cong P \oplus Q$. If any two of M, P, Q are finitely generated stably free projectives, then the third is a stably free projective. Combining this with the above argument, we obtain that P is free. ■

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